

## Singular Value Decomposition (SVD)

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \sigma \begin{pmatrix} u \\ v \end{pmatrix}$$

Singular Valued Decomposition, SVD, general linear maps

1. **Singular Value Decomposition**
2. General Linear Mappings
3. Linear Regression Revisited

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## Singular Valued Decomposition

- One of the most important results in linear algebra is that any matrix  $X$  can be decomposed as

$$X = USV^T$$

- If  $X$  is a  $m \times n$  matrix then  $U$  is a  $m \times m$  orthogonal matrix and  $V$  is an  $n \times n$  orthogonal matrix i.e.  $UU^T = U^TU = I$ ,  $VV^T = V^TV = I$
- And  $S$  is a  $m \times n$  matrix with non-zero diagonal elements  $S_{ii} = s_i$  known as the **singular values**
- Where did that come from?

## Eigen Decomposition

- Consider an arbitrary  $n \times m$  matrix  $X$ , and construct the  $(n + m) \times (n + m)$  symmetric matrix,  $B$ ,

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \sigma \begin{pmatrix} u \\ v \end{pmatrix}$$

$\begin{pmatrix} u \\ v \end{pmatrix}$  is an eigenvector of  $B$  with eigenvalue  $s$

- We observe that

$$\begin{aligned} Xv &= su \\ X^T Xv &= sX^T u = s^2 v \\ X^T u &= sv \\ XX^T u &= sXv = s^2 u \end{aligned}$$

## Eigenvectors

- Note that as  $\mathbf{X}\mathbf{v} = s\mathbf{u}$  and  $\mathbf{X}^T\mathbf{u} = s\mathbf{v}$  then

$$\mathbf{X}(-\mathbf{v}) = (-s)\mathbf{u} \quad \mathbf{X}^T\mathbf{u} = (-s)(-\mathbf{v})$$

if  $\begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix}$  is an eigenvector of  $\mathbf{B}$  with eigenvalue  $s$  then so is  $\begin{pmatrix} \mathbf{u} \\ -\mathbf{v} \end{pmatrix}$  with eigenvalue  $-s$

- If  $n < m$  then  $\mathbf{X}^T\mathbf{X}$  is not full rank so some eigenvalues are zero
- As a consequence  $m - n$  vectors exist such that  $\mathbf{X}\mathbf{v} = 0$
- The eigenvalues and eigenvectors are

$$n \times \left( s_i, \begin{pmatrix} \mathbf{u}_i \\ \mathbf{v}_i \end{pmatrix} \right) \quad n \times \left( -s_i, \begin{pmatrix} \mathbf{u}_i \\ -\mathbf{v}_i \end{pmatrix} \right) \quad m - n \times \left( 0, \begin{pmatrix} 0 \\ \mathbf{v}_k \end{pmatrix} \right)$$

## Matrix Decomposition

- Stacking the eigenvectors into a matrix

$$\begin{pmatrix} 0 & \mathbf{X} \\ \mathbf{X}^T & 0 \end{pmatrix} = \begin{pmatrix} \mathbf{U} & \mathbf{U} & 0 \\ \mathbf{V} & -\mathbf{V} & \mathbf{V}_0 \end{pmatrix} = \begin{pmatrix} \mathbf{U} & \mathbf{U} & 0 \\ \mathbf{V} & -\mathbf{V} & \mathbf{V}_0 \end{pmatrix} \begin{pmatrix} \Sigma & 0 & 0 \\ 0 & -\Sigma & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

- Since the vectors  $\begin{pmatrix} \mathbf{u}_i \\ \mathbf{v}_i \end{pmatrix}$  are eigenvectors of a symmetric matrix they form an orthogonal matrix if they are normalised.
- Multiply on the right by the transpose of the orthogonal matrix

$$\begin{pmatrix} 0 & \mathbf{X} \\ \mathbf{X}^T & 0 \end{pmatrix} = \begin{pmatrix} \mathbf{U} & \mathbf{U} & 0 \\ \mathbf{V} & -\mathbf{V} & \mathbf{V}_0 \end{pmatrix} \begin{pmatrix} \Sigma & 0 & 0 \\ 0 & -\Sigma & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{U}^T & \mathbf{V}^T \\ \mathbf{U}^T & -\mathbf{V}^T \\ 0 & \mathbf{V}_0^T \end{pmatrix}$$

## Normalisation Subtlety

$$\begin{pmatrix} 0 & \mathbf{X} \\ \mathbf{X}^T & 0 \end{pmatrix} = \begin{pmatrix} \mathbf{U} & \mathbf{U} & 0 \\ \mathbf{V} & -\mathbf{V} & \mathbf{V}_0 \end{pmatrix} \begin{pmatrix} \Sigma & 0 & 0 \\ 0 & -\Sigma & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{U}^T & \mathbf{V}^T \\ \mathbf{U}^T & -\mathbf{V}^T \\ 0 & \mathbf{V}_0^T \end{pmatrix}$$

- Multiplying out we have
- Now the vectors  $\mathbf{u}_i$  and  $\mathbf{v}_i$  form an orthogonal set as it satisfy
- But they are not normalised (since  $\begin{pmatrix} \mathbf{u}_i \\ \mathbf{v}_i \end{pmatrix}$  is normalised). If we define  $\tilde{\mathbf{U}} = \sqrt{2}\mathbf{U}$  and  $\tilde{\mathbf{V}} = \sqrt{2}\mathbf{V}$  we find

$$\mathbf{X} = 2\mathbf{U}\mathbf{S}\mathbf{V}^T \quad \mathbf{X}^T = 2\mathbf{V}\mathbf{S}\mathbf{U}^T$$

$$\mathbf{X}^T\mathbf{X}\mathbf{v} = s^2\mathbf{v} \quad \mathbf{X}\mathbf{X}^T\mathbf{u} = s^2\mathbf{u}$$

$$\mathbf{X} = \tilde{\mathbf{U}}\mathbf{S}\tilde{\mathbf{V}}^T \quad \mathbf{X}^T = \tilde{\mathbf{V}}\mathbf{S}^T\tilde{\mathbf{U}}^T$$

## SVD

- Any matrix,  $\mathbf{X}$ , can be written as  $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ 
  - $\mathbf{U}, \mathbf{V}$  are orthogonal matrices
  - $\mathbf{S} = \text{diag}(s_1, s_2, \dots, s_n)$
- $s_i$  can always be chosen to be positive and are known as **singular values**
- Singular value decomposition applies to both square and non-square matrices—they describe general linear mappings

## Finding SVD

- Most libraries will compute the SVD for you
- They can do this by choosing the smaller of two matrices  $\mathbf{X}\mathbf{X}^T$  and  $\mathbf{X}^T\mathbf{X}$  and then compute the eigenvalues
- The singular values are the square root of the eigenvalues (notice that  $\mathbf{X}\mathbf{X}^T$  and  $\mathbf{X}^T\mathbf{X}$  are both positive semi-definite so the eigenvalues will be non-negative)
- It can compute the  $\mathbf{U}$  matrix or  $\mathbf{V}$  matrix by multiplying through by  $\mathbf{X}$  or  $\mathbf{X}^T$  ( $\mathbf{U} = \mathbf{X}\mathbf{V}\mathbf{S}^{-1}$  and  $\mathbf{V} = \mathbf{X}^T\mathbf{U}\mathbf{S}^{-1}$ )
- In practice to perform PCA most people subtract the mean from their data and then perform SVD

## Economical Forms of SVD

- Often the rows or columns of the orthogonal matrices  $\mathbf{U}$  and  $\mathbf{V}$  that are not associated with a singular value are ignored

$$\begin{pmatrix} \mathbf{X} \\ \text{---} \end{pmatrix} = \begin{pmatrix} \mathbf{u} \\ \text{---} \end{pmatrix} \begin{pmatrix} \mathbf{S} \\ \text{---} \end{pmatrix} \begin{pmatrix} \mathbf{V}^T \\ \text{---} \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{X} \\ \text{---} \end{pmatrix} = \begin{pmatrix} \mathbf{u} \\ \text{---} \end{pmatrix} \begin{pmatrix} \mathbf{S} \\ \text{---} \end{pmatrix} \begin{pmatrix} \mathbf{V}^T \\ \text{---} \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{X} \\ \text{---} \end{pmatrix} = \begin{pmatrix} \mathbf{u} \\ \text{---} \end{pmatrix} \begin{pmatrix} \mathbf{S} \\ \text{---} \end{pmatrix} \begin{pmatrix} \mathbf{V}^T \\ \text{---} \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{X} \\ \text{---} \end{pmatrix} = \begin{pmatrix} \mathbf{u} \\ \text{---} \end{pmatrix} \begin{pmatrix} \mathbf{S} \\ \text{---} \end{pmatrix} \begin{pmatrix} \mathbf{V}^T \\ \text{---} \end{pmatrix}$$

- In Matlab these are obtained using

```
>> [U, S, V] = svd(X)
>> [U, S, V] = svd(X, 'econ')
```

## Outline

1. Singular Value Decomposition
2. **General Linear Mappings**
3. Linear Regression Revisited

$$\begin{pmatrix} 0 & \mathbf{X} \\ \mathbf{X}^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix} = \sigma \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix}$$

## General Matrix

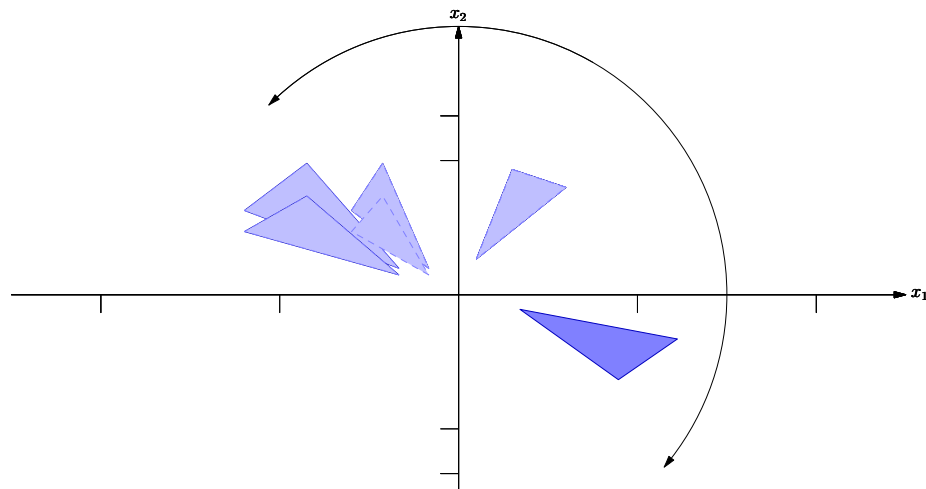
- Recall that we can compute the SVD for any matrix,  $\mathbf{X}$
- As matrices describe the most general linear mapping

$$\mathbf{v} \rightarrow \mathcal{T}[\mathbf{v}] = \mathbf{X}\mathbf{v}$$

- We can use SVD to understand any linear mapping
- Thus any linear mapping can be seen as a rotation followed by a squashing or expansion independently in each coordinate followed by another rotation

## Matrices

$$\mathbf{M} = \begin{pmatrix} -0.45 & 1.9 \\ -0.77 & -0.025 \end{pmatrix} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T = \begin{pmatrix} \cos(-175) & \sin(-175) \\ -\sin(-175) & \cos(-175) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75) & \sin(75) \\ -\sin(75) & \cos(75) \end{pmatrix}$$



## Determinants

- The determinant,  $|\mathbf{M}|$  of a matrix  $\mathbf{M}$  is defined for square matrices
- It describes the change in volume under the mapping
- Now for any two matrices  $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$
- Thus  $|\mathbf{M}| = |\mathbf{U}||\mathbf{S}||\mathbf{V}^T|$
- For an orthogonal matrix  $|\mathbf{U}| = \pm 1$  since  $\mathbf{U}\mathbf{U}^T = \mathbf{I} \Rightarrow |\mathbf{U}\mathbf{U}^T| = |\mathbf{I}| \Rightarrow |\mathbf{U}||\mathbf{U}^T| = 1$  or  $|\mathbf{U}|^2 = 1$
- Thus

$$|\mathbf{M}| = \pm |\mathbf{S}| = \pm \prod_i s_i$$

## Non-Square Matrices

- When the matrices are non-square then the matrix of singular value matrix will either
  - Squash some directions to zero
  - Introduce new dimensions orthogonal to the vector

$$\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T \quad \mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$$

- The rank of an arbitrary matrix is the number of non-zero singular values (also number of linearly independent rows or columns)

## Duality Revisited

- If  $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$  then

$$\begin{aligned} \mathbf{C} &= \mathbf{X}\mathbf{X}^T & \mathbf{D} &= \mathbf{X}^T\mathbf{X} \\ &= \mathbf{U}\mathbf{S}\mathbf{V}^T\mathbf{V}\mathbf{S}^T\mathbf{U}^T & &= \mathbf{V}\mathbf{S}^T\mathbf{U}^T\mathbf{U}\mathbf{S}\mathbf{V}^T \\ &= \mathbf{U}(\mathbf{S}\mathbf{S}^T)\mathbf{U}^T & &= \mathbf{V}(\mathbf{S}^T\mathbf{S})\mathbf{V}^T \end{aligned}$$

- If  $\mathbf{X}$  is an  $p \times m$  matrix then  $\mathbf{S}\mathbf{S}^T$  is a  $p \times p$  diagonal matrix with elements  $S_{ii}^2 = s_i^2$
- $\mathbf{S}^T\mathbf{S}$  is an  $m \times m$  matrix with elements  $S_{ii}^2 = s_i^2$
- $\mathbf{U}$  and  $\mathbf{V}$  are matrices of eigenvectors for  $\mathbf{C}$  and  $\mathbf{D}$
- The eigenvalues are  $\lambda_i = S_{ii}^2 = s_i^2$

## SS<sup>T</sup> and S<sup>T</sup>S

$$S = \begin{pmatrix} s_1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & s_2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & s_m & 0 & 0 \cdots & 0 \end{pmatrix}$$

$$S^T S = \begin{pmatrix} s_1^2 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & s_2^2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & s_m^2 & 0 & 0 \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 \cdots & 0 \end{pmatrix}$$

$$SS^T = \begin{pmatrix} s_1^2 & 0 & \cdots & 0 \\ 0 & s_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & s_m^2 \end{pmatrix}$$

## Having A Go

- It's really easy to verify this in MATLAB or OCTAVE

```
>> X = rand(3,2)
>> [U, S, V] = svd(X)
>> U*S*V'
>> U(:,1)'*U(:,2)
>> U'*U
>> U*U'
>> [Ua,L] = eig(X*X')
>> S*S'
```

- Test yourself!

## Outline

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- General Linear Mappings
- Linear Regression Revisited**

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \sigma \begin{pmatrix} u \\ v \end{pmatrix}$$

## Linear Regression

- Given a set of data  $\mathcal{D} = \{(x_i, y_i) | k = 1, 2, \dots, m\}$
- In linear regression we try to fit a linear model

$$f(x|w) = x^T w$$

- Which we fit by minimising the squared error loss

$$L(w) = \sum_{k=1}^m (f(x_k|w) - y_k)^2$$

## Matrix Form

- In matrix form we write  $L(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_m^\top \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}$$

- Then  $\nabla L(\mathbf{w}^*) = 0$  implies

$$\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \mathbf{X}^+ \mathbf{y}$$

- This is known as the pseudo-inverse

## Using SVD

- Using  $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$  then

$$\begin{aligned} \mathbf{X}^+ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ &= (\mathbf{V}\mathbf{S}^\top \mathbf{S}\mathbf{V}^\top)^{-1} \mathbf{V}\mathbf{S}^\top \mathbf{U}^\top \\ &= \mathbf{V}(\mathbf{S}^\top \mathbf{S})^{-1} \mathbf{V}^\top \mathbf{V}\mathbf{S}^\top \mathbf{U}^\top \\ &= \mathbf{V}(\mathbf{S}^\top \mathbf{S})^{-1} \mathbf{S}^\top \mathbf{U}^\top = \mathbf{V}\mathbf{S}^+ \mathbf{U}^\top \end{aligned}$$

- If  $m > p$

$$\mathbf{X}^\top = \begin{pmatrix} | & | & | & | & | & | & | & | \end{pmatrix}, \mathbf{S}^\top = \begin{pmatrix} s_1 & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & s_2 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & 0 & s_3 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & s_p & 0 & 0 & \cdots & 0 \end{pmatrix}$$

## Pseudo-Inverse of S

$$\mathbf{S}^\top \mathbf{S} = \begin{pmatrix} s_1^2 & 0 & \cdots & 0 \\ 0 & s_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & s_p^2 \end{pmatrix}, (\mathbf{S}^\top \mathbf{S})^{-1} = \begin{pmatrix} s_1^{-2} & 0 & \cdots & 0 \\ 0 & s_2^{-2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & s_p^{-2} \end{pmatrix}$$

$$\mathbf{S}^+ = (\mathbf{S}^\top \mathbf{S})^{-1} \mathbf{S}^\top = \begin{pmatrix} s_1^{-1} & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & s_2^{-1} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & 0 & s_3^{-1} & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & s_p^{-1} & 0 & 0 & \cdots & 0 \end{pmatrix}$$

## Ill-Conditioned Data Matrix

- Recall that

$$\mathbf{w}^* = \mathbf{X}^+ \mathbf{y} = \mathbf{V}\mathbf{S}^+ \mathbf{U}^\top \mathbf{y}$$

- If any of the singular values of  $\mathbf{X}$  are small then  $\mathbf{S}^+$  will magnify components in that direction
- Any errors in the target  $\mathbf{y}$  will be magnified
- This leads to poor weights

## Regularisation

- Consider linear regression with a regulariser

$$\begin{aligned}\mathcal{L}(\mathbf{w}) &= \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \eta \|\mathbf{w}\|^2 \\ &= \mathbf{w}^\top (\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I}) \mathbf{w} - 2\mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y}\end{aligned}$$

- Thus

$$\nabla \mathcal{L}(\mathbf{w}) = 2(\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I})\mathbf{w} - 2\mathbf{X}^\top \mathbf{y}$$

- and  $\nabla \mathcal{L}(\mathbf{w}^*) = 0$  gives

$$\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

## Regularisation Continued

- Using  $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$

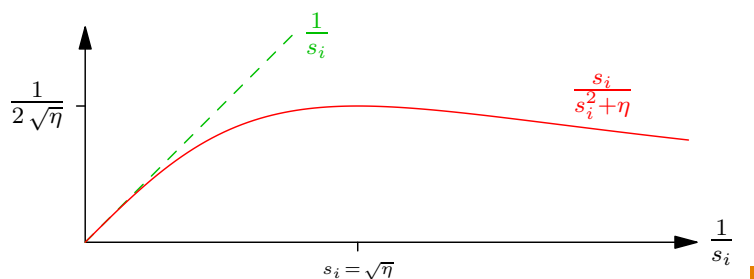
$$\begin{aligned}\mathbf{w}^* &= (\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y} \\ &= \mathbf{V}(\mathbf{S}^\top \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^\top \mathbf{U}^\top \mathbf{y}\end{aligned}$$

- where

$$(\mathbf{S}^\top \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^\top = \begin{pmatrix} \frac{s_1}{s_1^2 + \eta} & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \frac{s_2}{s_2^2 + \eta} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & 0 & \frac{s_3}{s_3^2 + \eta} & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{s_p}{s_p^2 + \eta} & 0 & 0 & \cdots & 0 \end{pmatrix}$$

## Effect of Regularisation

- Without regularisation if  $s_i = 0$  the problem would be ill-posed (even  $\mathbf{S}^+$  does not exist since  $s_i^{-1}$  would be ill defined) and if  $s_i$  is small then  $\mathbf{S}^+$  is ill conditioned
- Using  $\hat{\mathbf{S}}^+ = (\mathbf{S}^\top \mathbf{S} + \eta)^{-1} \mathbf{S}^\top$  instead of  $\mathbf{S}^+$  then



- Regularisation makes the machine much more stable (reduces the variance)

## Summary

- Any matrix can be decomposed as  $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$  where
  - $\mathbf{U}$  and  $\mathbf{V}$  are orthogonal (rotation matrices)
  - $\mathbf{S} = \text{diag}(s_1, \dots, s_n)$  is a diagonal matrix of positive singular values
- This describes the most general linear transform
- The transform exploits the duality between  $\mathbf{X}\mathbf{X}^\top$  and  $\mathbf{X}^\top \mathbf{X}$
- In linear regression the pseudo-inverse involves the reciprocal of the singular values, which can lead to poor generalisation
- Regularisation improves the conditioning of the “inverse” matrix